

# LOCAL SEARCH

**Hill Climbing • Simulated Annealing • Applications**

Russell & Norvig | Optimization-oriented search

**Searching for a good solution without keeping the entire path**

# Learning Objectives

- Understand the idea of local search and optimization problems.
- Explain the working of Hill-Climbing Search.
- Understand local maxima, plateaus and ridges.
- Explain how Simulated Annealing escapes local optima.
- Compare Hill Climbing and Simulated Annealing.
- Identify practical applications of local search.

# What Is Local Search?

Local search focuses on the current state and its neighbors.

## CLASSICAL SEARCH

Usually builds a path from initial state toward a goal and may retain a frontier of alternatives.

## LOCAL SEARCH

Keeps mainly the current state and moves to a neighboring state that may improve the objective.

**Key idea: “I have a candidate solution. Can I improve it by making a small change?”**

Useful when the final configuration matters more than the path used to reach it.

# Optimization View of Local Search

## STATE

One candidate solution.

## NEIGHBORS

Solutions obtained by small changes.

## VALUE / COST

Measures how good the candidate is.

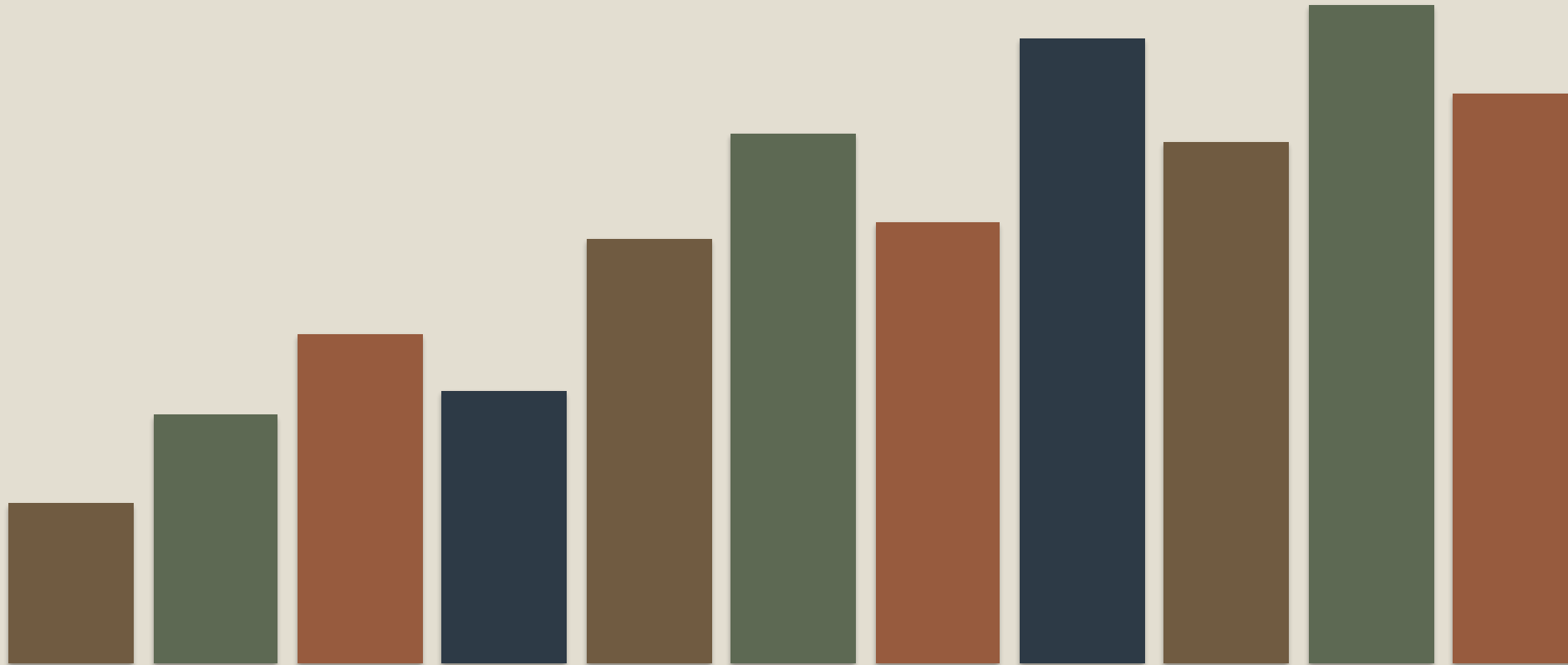
**Goal: move through the landscape toward a high-value state (or low-cost state).**

Examples: minimize conflicts, minimize travel distance, minimize production cost.

# A Simple Landscape

VALUE

HIGHER



LOW

A local search algorithm moves from one state to a neighboring state.

# Hill-Climbing Search

A greedy local optimization method.

## CORE RULE

Move to a neighboring state that improves the current state.

## INTUITION

“If my neighbor is better, move there.”

- Start with an initial candidate.
- Evaluate neighboring candidates.
- Move to a better neighbor.
- Repeat until no neighbor is better.

# Hill Climbing — Step by Step

## 1. START

Current state  
value = 40

## 2. CHECK

Neighbors:  
45, 38, 42

## 3. MOVE

Choose 45  
because  $45 > 40$

## 4. REPEAT

Continue until no neighbor  
improves the value.

**$40 \rightarrow 45 \rightarrow 52 \rightarrow 57 \rightarrow 57$**

At 57, suppose every neighboring state has value  $\leq 57$ .

**Then Hill Climbing stops.**

# Problem with Hill Climbing: Local Maximum

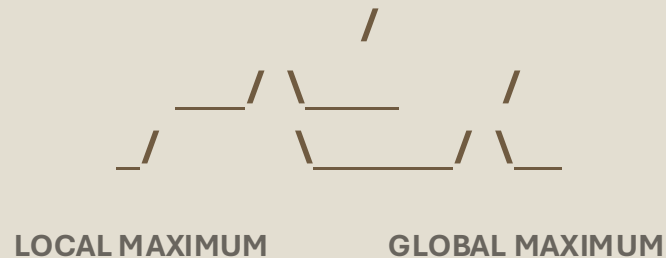
## GLOBAL MAXIMUM

The best solution in the entire search landscape.

## LOCAL MAXIMUM

A state better than all its immediate neighbors, but not the best overall.

Hill climbing can stop here:



# Other Hill-Climbing Problems

## PLATEAU

A flat region where neighboring states have the same value. The algorithm may have no obvious uphill move.

## RIDGE

A narrow region where the best direction may require a sequence of moves that are not immediately better.

## LOCAL MAXIMUM

All immediate neighbors look worse, so the algorithm stops too early.

**These limitations motivate methods that can sometimes accept a worse move.**

# Ways to Improve Hill Climbing

- Random-restart hill climbing: run hill climbing from several random initial states.
- Stochastic hill climbing: choose randomly among improving moves.
- First-choice hill climbing: examine neighbors until an improving move is found.
- These methods can reduce sensitivity to the initial state.
- Simulated Annealing goes further by sometimes accepting worse moves.

# Simulated Annealing

A local search method inspired by the cooling of materials.

## CORE IDEA

Initially, the algorithm can accept some worse moves to escape local maxima.

## AS TEMPERATURE FALLS

The algorithm becomes less willing to accept worse moves and settles into a good region.

**HIGH TEMPERATURE** → more exploration

**LOW TEMPERATURE** → more exploitation

**Goal: escape poor local optima early, then stabilize.**

# How Simulated Annealing Works

## 1. CURRENT

Start with a candidate state.

## 2. NEIGHBOR

Randomly choose a neighboring state.

## 3. COMPARE

Better → usually accept.  
Worse → may still accept.

## 4. COOL

Reduce temperature and continue.

**Unlike basic hill climbing, a worse move is NOT automatically rejected.**

**That ability helps the search jump out of local maxima.**

# The Acceptance Probability

## WORSE MOVE

Suppose the new state is worse by  $\Delta E$ .

## PROBABILITY

A common rule uses:

$$P = e^{(-\Delta E / T)}$$

where  $T$  = temperature.

**High  $T \rightarrow$  higher chance of accepting a worse move.**

**Low  $T \rightarrow$  lower chance of accepting a worse move.**

You do not need to calculate the probability to understand the core idea.

# Hill Climbing vs Simulated Annealing

## HILL CLIMBING

Better move → accept

Worse move → reject

Risk: gets stuck at local maximum.

## SIMULATED ANNEALING

Better move → accept

Worse move → sometimes accept

Temperature gradually decreases.

**Hill Climbing = greedy local improvement**

**Simulated Annealing = exploration first, stabilization later**

# Local Search Application: 8-Queens

## PROBLEM

Place 8 queens on a chessboard so that no two queens attack each other.

## LOCAL SEARCH VIEW

A state = one queen in each column. A neighbor = move one queen to another row.

**Objective function example: minimize the number of attacking queen pairs.**

**Hill climbing repeatedly moves toward fewer conflicts.**

# Local Search Applications

## SCHEDULING

Assign classes, exams, machines or jobs while minimizing conflicts and idle time.

## ROUTE / TSP

Find a short or low-cost route through many locations.

## RESOURCE ALLOCATION

Allocate limited resources while minimizing cost or maximizing benefit.

## LAYOUT DESIGN

Place components to reduce wiring, distance or congestion.

## NETWORK OPTIMIZATION

Improve routing, capacity or connectivity objectives.

## ENGINEERING DESIGN

Optimize parameters for performance, cost, weight or energy.

# Why Use Local Search?

- It can work with very large or complex state spaces.
- It often stores only a small amount of search information.
- It is suitable when we care about the final configuration rather than the exact path.
- It is useful for optimization problems where finding a perfect solution may be expensive.
- Stochastic methods can explore different regions and reduce the chance of getting trapped.

# Exam / Viva Focus

- What is local search?
- How does Hill Climbing choose the next state?
- What is a local maximum?
- What is a plateau? What is a ridge?
- Why can Hill Climbing fail to find the global optimum?
- How does Simulated Annealing escape local maxima?
- What is the role of temperature  $T$ ?
- Give three real-world applications of local search.

# FINAL TAKEAWAY

**LOCAL SEARCH = improve a candidate solution using its neighbors**

## Hill Climbing

→ always move to a better neighbor

## Simulated Annealing

→ sometimes accept worse moves

## Applications

→ scheduling, routing, design,  
allocation

**Think: Hill Climbing exploits. Simulated Annealing explores, then settles.**